

## GPN1 □ - 40.5kV Removable AC Metal-clad Enclosed Switchgear

### Summary

GPN1-40.5kV removable AC metal-clad switchgear (short for switchgear as below) applies to 40.5kV, 3 phase, AC and 50/60Hz electrical power network for receiving and distributing power energy and also for control, monitor and protection. It is applicable in general power system and occasions with frequent operation. The product conforms to GB 3906 "3~35kV AC Metal Enclosed Switchgear", GB/T 11022 "The common technical clauses of High Voltage Switchgear and controlgear Standard", DL/T 404-1997 "Order Technical term of Indoor AC High Voltage Switchgear" and IEC-298 "1~52kV AC Metal Enclosed Switchgear and Controlgear standard".



### Ambient condition

Ambient temperature: -25°C ~ +45°C , daily average: ≤ 35°C  
 Altitude: 1000m(Standard); can up to 4500m for special ordering;  
 Relative humidity: daily average ≤ 95%, monthly average ≤ 90%;  
 Earthquake intensity: ≤ 8 degree;  
 Applicable occasions should be free from corrosives, inflammables and vapor.

### Model and its meaning

|    |   |   |   |   |      |   |   |      |   |      |                                                  |
|----|---|---|---|---|------|---|---|------|---|------|--------------------------------------------------|
| GP | N | 1 | □ | - | 40.5 | / | T | 2000 | - | 31.5 |                                                  |
|    |   |   |   |   |      |   |   |      |   |      | Rated short circuit breaking current (kA)        |
|    |   |   |   |   |      |   |   |      |   |      | Rated current (A)                                |
|    |   |   |   |   |      |   |   |      |   |      | Spring operating mechanism                       |
|    |   |   |   |   |      |   |   |      |   |      | Rated voltage (kV)                               |
|    |   |   |   |   |      |   |   |      |   |      | A: Middle-plated; B: floor type; C: VD4/HD4 type |
|    |   |   |   |   |      |   |   |      |   |      | Air insulated                                    |
|    |   |   |   |   |      |   |   |      |   |      | Indoor removable metal-clad                      |
|    |   |   |   |   |      |   |   |      |   |      | Company feature code                             |

### Product feature

The switchgear is an assembly unit and the circuit breaker adopts handcart floor model. It owns simply change and well exchange after equipping with advanced composite insulation vacuum circuit breaker. Lead screw nut propulsion mechanism is installed in the handcart frame for preventing from fault operating and damaging propulsion mechanism, the handcart can be remove easily; All of the operation steps can be carried out at closing condition; It is compulsory locking among main switch, handcart and panel door to meet five protection; wide space for connecting several cables in cable cubicle. Earthing switch is used in earthing and loop short circuit, protection degree of enclosure is IP3X, IP2X under opening condition.

### Technical specification

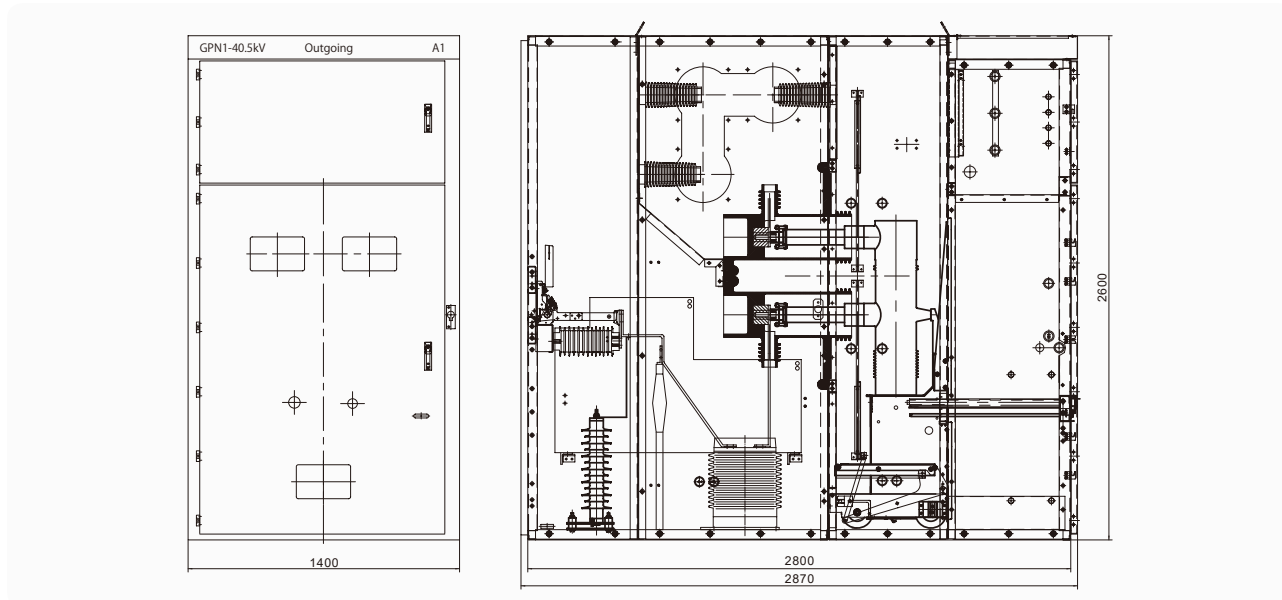
| No. | Item                                      |                                                          | Unit | Data                                                            |
|-----|-------------------------------------------|----------------------------------------------------------|------|-----------------------------------------------------------------|
| 1   | Rated voltage                             |                                                          | kV   | 36/38/40.5                                                      |
| 2   | Rated current                             |                                                          | A    | 1250, 1600, 2000, 2500                                          |
| 3   | Rated frequency                           |                                                          | Hz   | 50/60                                                           |
| 4   | Rated short circuit breaking current      |                                                          | kA   | 25, 31.5                                                        |
| 5   | 4s rated short time withstand current     |                                                          | kA   | 25, 31.5                                                        |
| 6   | Rated short circuit making current (peak) |                                                          | kA   | 63, 80                                                          |
| 7   | Rated peak withstand current(peak)        |                                                          | kA   | 63, 80                                                          |
| 8   | Insulation level                          | Rated lightning impulse withstand voltage                | kV   | 185 (across open contacts: 215)                                 |
|     |                                           | Main circuit 1min power frequency withstand voltage      | kV   | 95 (across open contacts: 110)                                  |
|     |                                           | Auxiliary circuit 1min power frequency withstand voltage | kV   | 2                                                               |
| 9   | Outline dimension(W)*(D)*(H)              |                                                          | mm   | 1400/2800/2600 (GPN1 type)<br>1200/2600/2400 (for VD4/HD4 type) |
| 10  | Protection degree                         |                                                          |      | IP3X (compartment IP2X)                                         |
| 11  | Weight                                    |                                                          | kg   | 1000-1850 (GPN1 type)<br>850-1850 (For VD4/HD4 type)            |

## ■ Structure feature

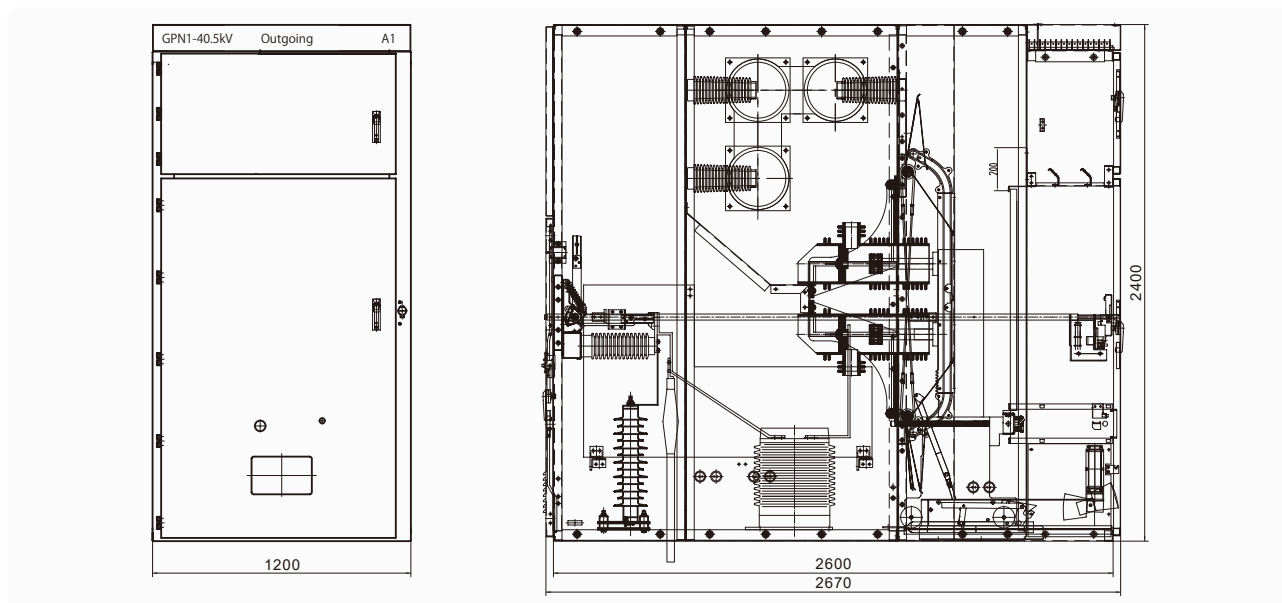
The switchgear divides into enclosure and handcart, the enclosure is made of aluminum-zinc plated steel sheet after processed by CNC machine and multi-bending, then is assembled with bolts. So it has strong mechanical strength and can guarantee the neatness and good appearance. It consist of relay compartment, handcart compartment, cable compartment, protection degree is IP2X when CB handcart is on opening and testing position.

The switchgear is metal-clad removable type, main circuit adopts composite insulation processing, emergency spreading to other parts. CT and earthing switch are mounted in cable compartment and busbar compartment. The protection degree of enclosure is IP3X, the protection and large space for several cables. The structural design of new completely insulated VCB or SF<sub>6</sub> CB and spring operating mechanism is integrated console model with such excellence of well exchange and simply change.

## ■ Outline dimension of the switchgear



GPN1-40.5 Structure drawing of standard type



GPN1(C)-40.5 Structure drawing for VD4/VH4 type

Note:

1. Motorized feeding for the VCB is available. Any special additional requirement of application, please contact us for technical disclosure.
2. Here only show the outgoing schematic for reference. For the other schematic, Please consult with GP for more details.